Appearance-based Facial Detection for Recognition

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***Abstract*—This paper propose an automatic method for facial features detection and then the image quality improvement methods to increase the rate of good recognition of a classifier based on appearance.**

***Keywords* - facial feature detection; vignetting; triangle face estimation.**

I. Introduction

The development of the types of embedded systems have led to improved methods that enable the biometric authentication, the video surveillance or the access control; all these applications being based on facial recognition algorithms. Currently, the facial recognition algorithms can be classified into two broad categories: (1) photometric or based on aspect, (2) geometric or based on features and models.

The recognition methods based on aspect enable the representation of objects as a series of different aspects of the same object. An image is regarded as a multidimensional vector, so that many of the algorithms use statistical methods of analysis of the distribution of objects in the vectorial space to obtain an effective way of representing the objects according to the application.

The face recognition algorithms based on models are based on the construction of a human face model, able to capture the variations of the faces. The a priori knowledge about the human face is used to build the models.

The study in this work aims to determine those procedures of digital image processing that enable the increase of the rate of human face recognition through techniques of improving the quality of the image and through techniques of detection of the specific regions - eyes, mouth, which allow for adjustments on the images, thus it introduces a classifier based on appearance, to increase the rate of good recognition.

II. The Evaluation Of The Performance

In order to evaluate the performance of the proposed algorithms, we built a database containing a sufficient series of potential contexts of the subjects that was as general as possible so as to obtain performance that was as relevant and explicit as possible. The database contains 1.068 images from 162 subjects depicted in a number of artistic images at different resolutions, orientations or lighting conditions and they may present occlusions of the entire surface of the face caused by predominant shadows or by hairstyle.

In order to assess the quality of the images, special importance has been given to the relationship between the recognition rate and the quality of the images under analysis, as this may lead to the selective application of certain corrections that can determine the improvement of the performance of the classifier. The quality of the image was assessed by analyzing the average brightness and by the threshold criterion of the Weber contrast [1], [2].

III. The Algorithm For Locating The Facial Features

The face detection was made based on the facial features most commonly used in such applications, namely the eyes and the mouth.

A. Detecting and locating the eyes

For locating the candidate eyes based on the color map in the YCbCr space [4], [5], on the symmetry map [6] and on the image intensity [3] a single I_E criterion was defined, according to (1).

$$I_E(x,y) = E(x,y)^{k_E} + [1 - S(x,y)]^{k_S} + [1 - I(x,y)]^{k_I}, \quad (1)$$

where k_E, k_S, k_I are the weights associated with the eye detection map, $E(x,y)$, to the symmetry map, $S(x,y)$ and to the image intensity, $I(x,y)$; k_E, k_S and k_I are higher than one. In equation (1), all the maps and the image intensity have values that are normalized in the interval $[0,1]$.

In order to remove the local minimum values that are very close in terms of distance, the I_E is filtered by means of spatial convolution with a Gaussian two-dimensional window W.

$$I_E^{filt} = I_E * W. \quad (2)$$

The region corresponding to a potential eye in the analyzed image will have a high value of the detection map $E(x,y)$ and a low value of the symmetry map $S(x,y)$ and of the intensity of the image $I(x,y)$. Due to the fact that the last two parameters have low values, these are reversed in the relation of the I_E criterion. A maximum of the I_E criterion will correspond to a maximum of probability for locating an eye at the associated coordinate (x,y).

The pixels candidate as potential eyes in the image represent the local minimum values and can be determined by means of the Watershed transform [7], [8]. This allows for the determination of the regions containing a single local minimum value. Therefore, we initially determined all those regions containing local minimum values, we found out these minimum values and retained the first N=2 values in descending order in relation to the module "Fig. 1".

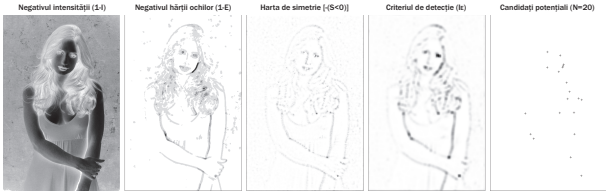

Figure 1. The result for determining the potential candidates for eye detection (the negative of the intensity - $(1 - I)$; the negative of the map of the eyes - $(1 - E)$, the symmetry map - (S<0), the detection criterion - (I_E), the candidate eyes - (N=20))

B. Detecting and locating the mouth

Determining the candidate regions for detecting and locating the mouth is based on the fact that this type of feature has a strong red component and a small amount of blue as compared to the other facial regions. Therefore, in the mouth region the C_r chrominance component is higher than the C_b component and has a reduced response of the C_r/C_b parameter, as well as a higher response for C_r^2. One can build the map M associated to the candidate regions for the mouth [4].

$$M = C_r^2 \cdot (C_r^2 - \eta \cdot C_r/C_b)^2 \tag{3}$$

for which:

$$\eta = 0.95 \cdot \frac{\frac{1}{n}\sum_{(x,y)\in F} C_r(x,y)^2}{\frac{1}{n}\sum_{(x,y)\in F} C_r(x,y)/C_b(x,y)}, \tag{4}$$

where C_r^2 and C_r/C_b are normalized in the domain $[0, 255]$, n is the number of pixels in the face mask F, and η is estimated as the ratio between the average values of the C_r^2 and C_r/C_b components.

Based on the facial features, one can determine those regions corresponding to a face pattern in terms of their spatial arrangement. One can use the fact that the eyes and mouth are displayed in the form of a triangular pattern [9], [10], [11] whose positions and distances follow the same rules. The detected region corresponds to the face and will be used by a classifier based on appearance.

C. The results of the detection

Initially, the correct detection of the facial features - the eyes and the mouth - was evaluated and then we tried to apply some selective corrections in order to improve the rate of good recognition. The results of the *eye detection* in the images of the database were analyzed according to different configurations of the algorithm: the height of the re-scaled images (H), the symmetry parameters (S and K), the threshold value of the Weber contrast (C), the gamma correction (γ), and the intensity correction (I).

In order to modify the resolution of an image (reducing its size) we use the following rule. Considering a maximum resolution H of pixels, the images that will have the height lower than H will remain unchanged, following that, for images with higher height, they will be rescaled by bilinear interpolation to the value of H. Height is considered as the dimensionality criterion because, for the images containing faces, the W/H ratio is generally lower than one. Thus, maintaining a constant height for all the faces in the database is possible.

The radial dimension of the symmetry elements (S) enables the search for symmetrical elements at various levels of resolution within the image. The radiality parameter (K) enables the introduction of a symmetry tolerance in terms of the deformation occurred in the shape of the eyes and caused by different angles or partial occlusions of the face.

The gamma correction aims at modifying the contrast of those images which have insufficient contrast, according to the Weber contrast criterion. The evaluation was made by applying a gamma correction to those images that had a negative Weber contrast. As a result of better detection for eye location, based on the tests that were carried out, one can notice that "Table I":

- the optimal dimensions obtained are H=325 pixels and H=350 pixels;
- the less strict the radiality parameter is (K>1, the greater the number of positive results;
- the gamma correction does not always have a positive effect. One can infer an algorithm to determine the optimal threshold value of the Weber contrast in order to apply the gamma correction and the brightness correction.

The results of the *mouth detection* from the database images were analyzed according to the same configurations of the algorithm, by eliminating the symmetry parameters and we obtained higher values than in the case of the eye detection. Unlike the eyes, which have a rounded, circular shape, the mouth has an elongated thin shape no matter whether it is closed or open. Therefore, it can not be determined by elements of symmetry detection. The percentages obtained are maximum for similar resolutions, H=300 pixels and H=325 pixels with detection values of 87.17% and the gamma correction does not significantly influence the results "Table II".

Combining the mouth and the eyes detection allows for the determination of good detection rates that are within the range 52.72% -56.18%.

If we take into account the situation where a single eye is detected (the second eye can be detected by symmetry), the percentage increases between 73.32% and 75.28%, with maximum resolution values of $H = 350$ pixels. The previous results, although with normal values for the types of images in the database, can be improved by taking into account the context of the analyzed images. The images that contain useful information have, in most situations, the objects positioned centrally. Therefore, the corners of the images are of no interest unless in terms of the background. Thus, the images containing human faces can be masked to select only the regions with the highest probability of interest.

IV. Techniques For Improving The Results

Based on previous observations, one can apply techniques that can lead to improved outcomes, namely:

- the application of ellipsoidal vignetting which would enable the elimination of the disturbing elements at the extremities of the image;

Table I
RESULTS OF THE DETECTION OF THE EYES WITH THE FIRST TWO MAXIMUM SYMMETRY VALUES

Test	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10
global	78.09%	80.24%	80.43%	79.96%	79.82%	79.87%	79.17%	79.17%	77.29%	77.06%
2 eyes	66.01%	67.79%	68.45%	68.26%	66.85%	66.67%	66.01%	66.20%	63.86%	63.76%
1 eyes	24.16%	24.91%	23.97%	23.41%	25.95%	25.94%	26.31%	25.94%	26.87%	26.59%
0 eyes	9.83%	7.20%	7.58%	8.33%	7.21%	7.40%	7.68%	7.87%	9.27%	9.64%
H	original	$\leq$325 px.	$\leq$350 px.	$\leq$400 px.	$\leq$325 px.	$\leq$350 px.	$\leq$325 px.	$\leq$350 px.	$\leq$325 px.	$\leq$350 px.
S	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7 9]	[1 3 5 7]	[1 3 5 7]
K	2	2	2	2	1	1	1	1	1	1
C	–	–	–	–	–	–	–	–	–	–
γ	–	–	–	–	–	–	0.75	0.75	0.90	0.90
I	–	–	–	–	–	–	–	–	–	–

Table II
RESULTS OF THE DETECTION OF THE MOUTH

Test	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11
mouth	86.99%	87.17%	87.17%	87.08%	87.08%	87.17%	87.08%	87.17%	87.08%	87.17%	87.08%
H	original	$\leq$300 px.	$\leq$325 px.	$\leq$350 px.	$\leq$400 px.	$\leq$325 px.	$\leq$350 px.	$\leq$325 px.	$\leq$350 px.	$\leq$325 px.	$\leq$350 px.
C	–	–	–	–	–	–	–	–	–	–	–
γ	1	1	1	1	1	0.90	0.90	0.80	0.80	0.70	0.70
I	–	–	–	–	–	–	–	–	–	–	–

- the application of similar transformations to estimate the position of an eye on the basis of the coordinates of the mouth and of the other eye.

A. Ellipsoidal vignetting

The ellipsoidal vignetting involves the elimination of the information content located outside the ellipse present in the image of W and H dimensions. The ellipsoidal mask will be described by the relationship:

$$\left(\frac{x - H/2}{kH/2}\right)^2 + \left(\frac{y - W/2}{kW/2}\right)^2 < 1, \tag{5}$$

where k is a dimensional parameter with values in the interval $[0, 1]$. In the case where k=1, the ellipse is tangent to the edges of the image, while the value of k tends to 0, the ellipse decreasing in the center of the image.

Where the ellipsoidal mask is applied, the results of the detection of the facial features are significantly improved. For the detection of the eyes and mouth, the best results are obtained for low resolutions of the image (H=250 pixels), which brings a new advantage, that of the rapidity of the calculation. The percentage values of the detection are, for the eye, of up to 86.89% and for the mouth of maximum 95.88%. The combined results confirm the previous observation, the obtained percentages being higher than those obtained without the application of the ellipsoidal vignetting. The cumulative percentages of detection of the mouth and of at least one eye have values between 89.61% and 93.08%.

B. Affine transformations applied to images

The percentage of good detection of the mouth is high so that for facial features detection, it is indicated to determine first the position of the mouth and then to attempt to determine the position of the eyes. In the situations in which one of the eyes can not be detected or is erroneously detected, its position can be extrapolated starting from the distance between the mouth and the correctly detected eye "Fig. 2".

Based on the statistical tests performed on the images from the generated database and on the anthropometric relations, the coordinates of the mouth and eyes are validated if the following conditions are met simultaneously:

$$\begin{aligned} 0.8 \leq \quad & D_M/D_E \quad \leq 1.6; \\ 35 \leq \quad & \alpha_{ME} \quad \leq 65. \end{aligned} \tag{6}$$

where D_E is the distance between the two eyes, D_M is the distance from the mouth to the middle of the D_E segment, and α_{ME} is the angle formed by the lines obtained by joining the mouth with each eye.

Initially, one needs a criterion for the selection/validation of an eye, for confirming the fact that, in addition to validating the triangle of the face, the point of the eye corresponds to a real eye in the image. To this end, one analyzes the nuance histogram within a disk around the undetected point. The size of this disc, in general, will have the size equal to a quarter of the distance between the two eyes "Fig. 3". Starting from the fact that any eye will implicitly have a nuance of white, one looks for the confirmation of this for each point detected as eye. First, based on the confirmed results of the detection of the eyes, one builds the histogram model of the color of the eye in the disc around the undetected points. The calculus of the white coefficient in the analyzed disk is performed as the weighted sum of the pixels that cumulatively meet the following conditions:

$$R > 200 \ \& \ G > 170 \ \& \ B > 170 \tag{7}$$

and the sum of the differences between the primary colors is not greater than 100:

$$|R - G| + |R - B| + |G - B| < 100. \tag{8}$$

Because the eye is vascularized, the R component of the color of the eyeball has a value that is greater than that of the components G and B.

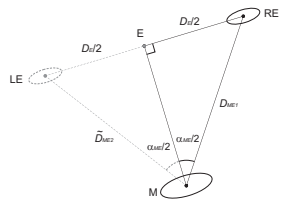

Figure 2. The estimation of the position of an undetected eye by means of the construction of the triangle of the face

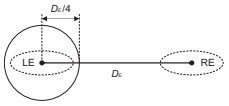

Figure 3. The circular region for determining the histogram of the color of the eye

If the face is rotated by a certain angle in the image plane, the gradient can be estimated starting from the best detection of the mouth. For the application of the estimation of the position of the undetected eye, one determines first the initial gradient of the face. Next, one determines the position of the undetected eye based on the un-rotated triangle, followed, by the calculation of the final position by rotating the triangle of the face with the gradient, considering the point corresponding to the mouth as the origin of the rotation. The necessary transformations for the point (X_{E2}, Y_{E2}) are shown 9-11.

$$\begin{aligned} X'_{E1} &= X_M - D_{ME}\cos\Delta\alpha \\ Y'_{E1} &= X_M - D_{ME}\sin\Delta\alpha \\ X'_{E2} &= X_M - D_{ME}\cos\Delta\alpha \\ Y'_{E2} &= X_M + D_{ME}\sin\Delta\alpha \end{aligned} \tag{9}$$

$$\begin{bmatrix} X''_{E2} \\ Y''_{E2} \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\Delta\alpha & -\sin\Delta\alpha & 0 \\ \sin\Delta\alpha & \cos\Delta\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X'_{E2} - X_M \\ Y'_{E2} - Y_M \\ 1 \end{bmatrix} \tag{10}$$

$$\begin{bmatrix} X_{E2} \\ Y_{E2} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & X_M \\ 0 & 0 & Y_M \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} X''_{E2} \\ Y''_{E2} \\ 1 \end{bmatrix} \tag{11}$$

where D_{ME} is the distance from the point of the mouth to the point of the detected eye and $\Delta\alpha$ is the difference between the determined angle of the segment D_{ME} and the average value 22.5ï¿œ of the $\alpha_{ME}/2$ which is determined experimentally.

The results obtained by means of the correction of the position of the eyes are superior. One obtains good detection rates between 82.30% (image without re-scaling) and 84.46% (image with H=325 pixels).

Moreover, the result of the combined detection of the eyes and mouth (the triangle of the face) is higher than that obtained in the absence of correction "Table III".

V. Conclusion

The paper has proposed a series of image processing techniques meant to improve the quality of images showing human faces, in order to use them in a classifier based on appearance. Images were analyzed from the point of view of information by establishing two types of facial features, the eyes and the mouth, respectively, based on mouth detection maps and on symmetry maps, proposing the implementation of a new method of using the Watershed transform which would extract the associated coordinates.

During a first stage we sought to determine a quantifier that would enable the a priori analysis of the quality of an image. Among the parameters that were used, the most useful was the Weber contrast because it takes into account both the region of interest and the background image in its definition, which is the situation encountered in the recognition of the face. On the other hand, the other contrast features are global values, without a direct relationship with the information content of the image under analysis. Subsequently, we determined the optimal level of resolution of the images in the database for which the application of the detection maps has the maximum result (the image height being between 325 and 350 pixels). The application of the ellipsoidal vignetting made it possible to eliminate from the picture the factors that are potentially disturbing for the detection of facial features. The results obtained in this case were higher than those without the application of vignetting.

In some cases, the result of the detection may contain some errors, like the false detection of an eye or of both eyes. For the first situation, we proposed a technique for the correction of the result of detection by determining the triangle like appearance of the face (composed of the two eyes and mouth), by determining a correctly detected eye by means of a detection mask and, based on this, of the point of the mouth as well, the estimation of the position of the eye undetected in the initial phase. The detection rates were higher when we applied the proposed method, increasing up to 82.21% as compared with 72.66% in the initial case.

Table III
The rate of detection of both eyes and of the mouth for the situation with and without the correction of the position

Normal	73.20%	72.66%	72.38%	71.72%	70.13%
Correction	81.18%	82.21%	81.37%	80.81%	79.12%
H	$\leq$300 px.	$\leq$325 px.	$\leq$350 px.	$\leq$400 px.	original

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